

Dictionary: The Dictionary is a type of Array where the contents consist of key and value. Here the key is used in place of the numeric index in order to identify an element.

Indexed Array: an indexed array can store any element that is included in the number element. Here the individual elements are identified by using their number or index. These Indexed Arrays can use a 32-bit unsigned integer for the index number.

Here we should always remember that the maximum size of an indexed Array is $2^{32} - 1$ or 4,294,967,295. We will inevitably face a run time error if we attempt to create an Array that is larger than this maximum size. Now let us look at the example below:

Example:

// Use Array constructor.

```
var myArray:Array = new Array();  
myArray.push("first");  
myArray.push("second");  
myArray.push("third");  
trace(myArray); // output: first,second,third
```

// Use Array literal.

```
var myArray:Array = ["first", "second", "third"];  
trace(myArray); // output: first,second,third
```

Element: elements are the items in an array. We will discuss elements in detail in the next section.

Multidimensional Array: a multidimensional array includes the items that are usually themselves identified as arrays and not as single values. This is an array of arrays. We will discuss multidimensional arrays in detail in the last section of this unit.

6.2 The Anatomy of an Array

An Array is composed of two things:

- Array elements
- Index number